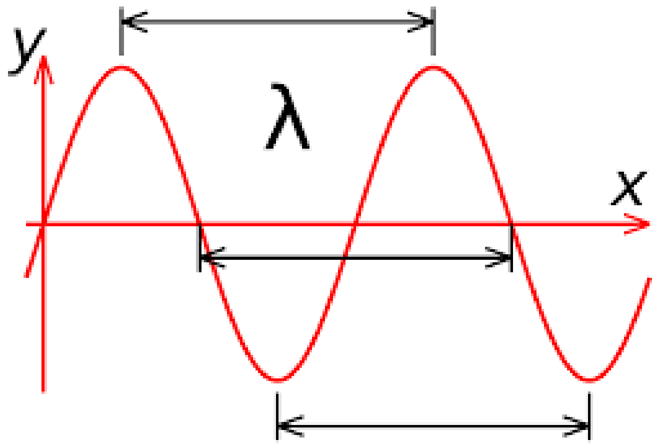
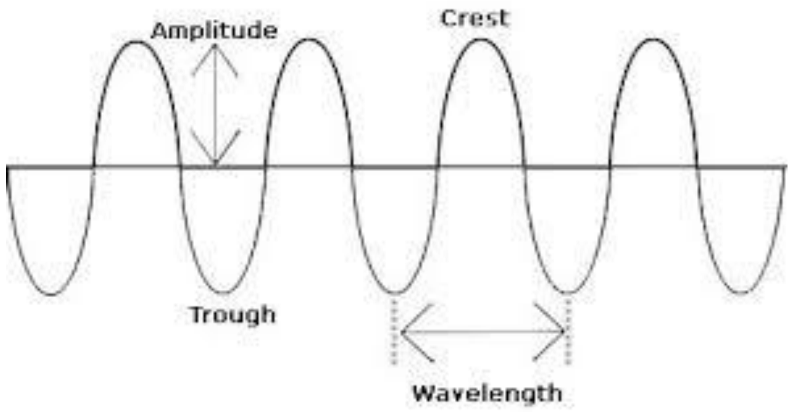
OPTICAL MEASURING INSTRUMENTS

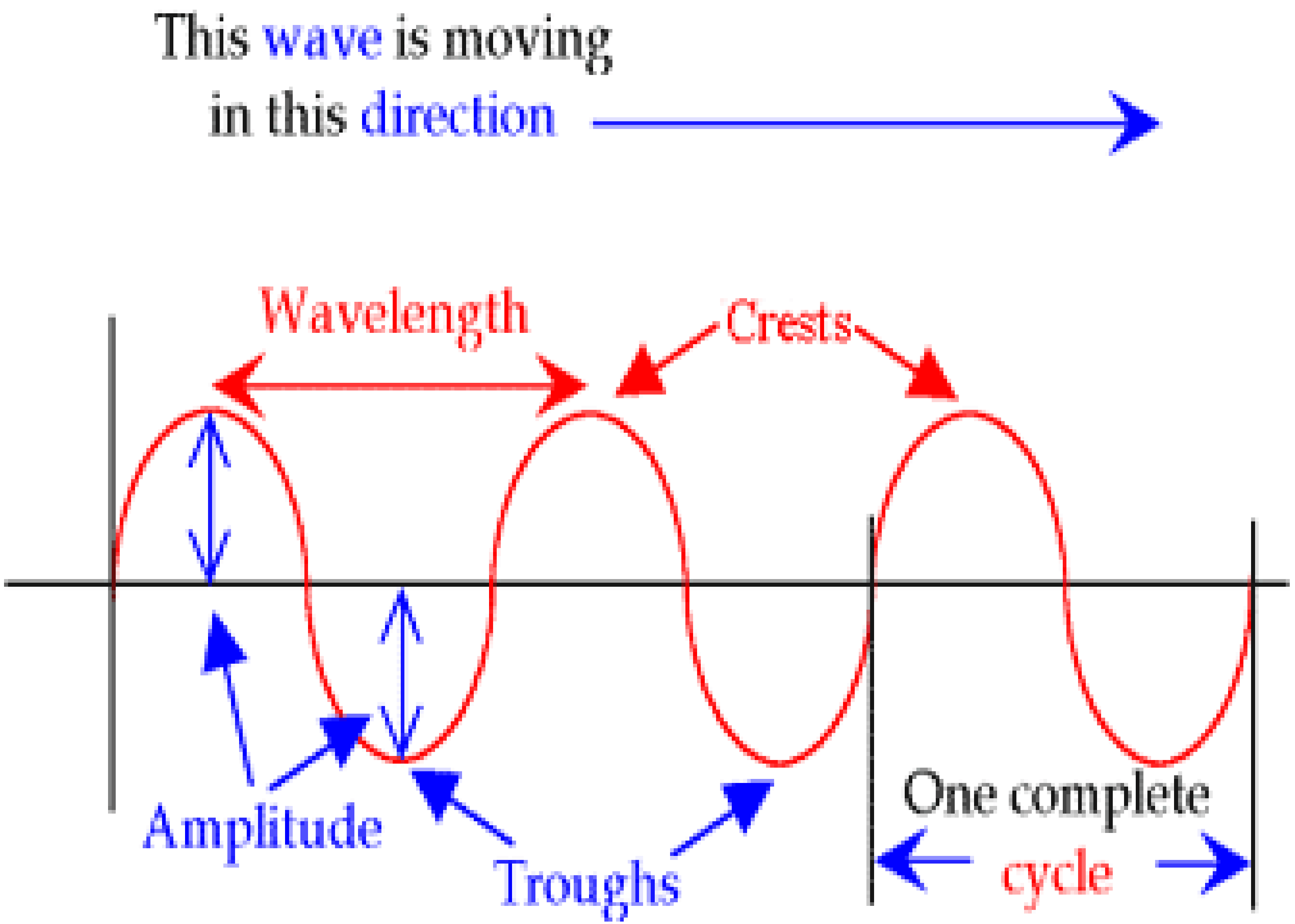
Introduction

Nature of light

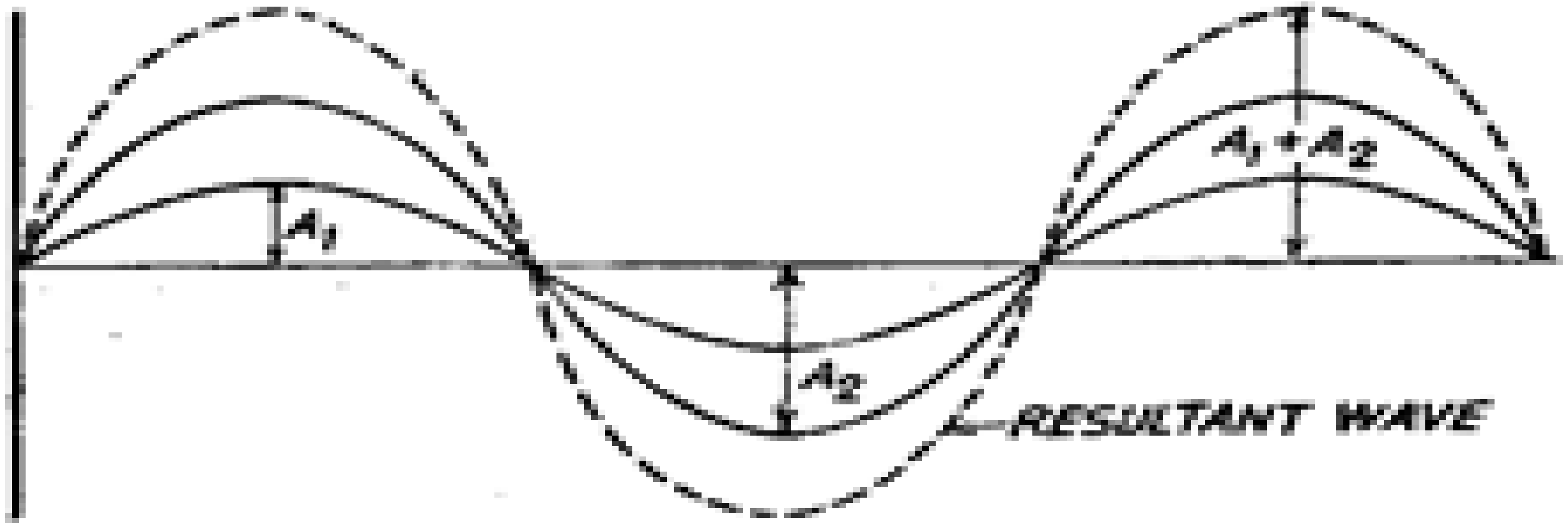
- Electromagnetic wave of sinusoidal form





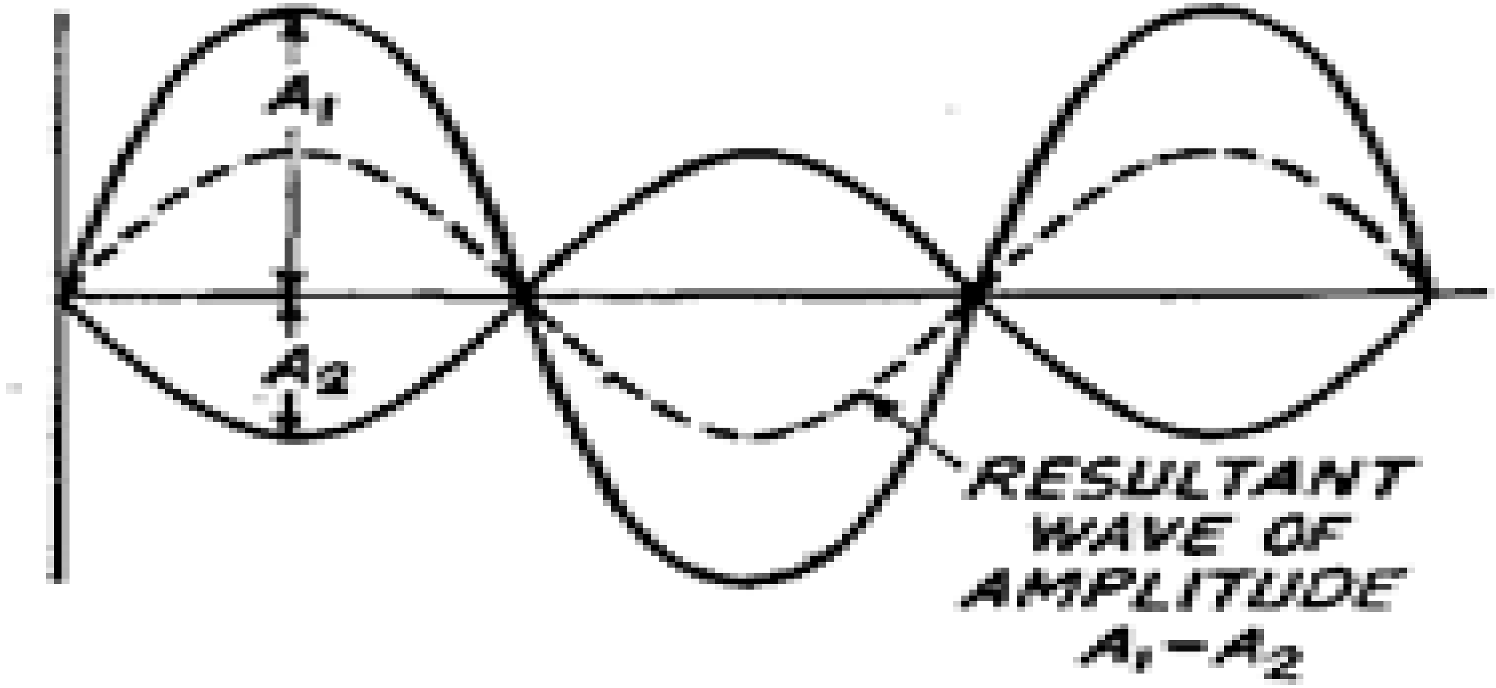

This wave is moving
in this direction
Wavelength
Crests
Amplitude
Troughs
One complete
cycle

interference of two rays



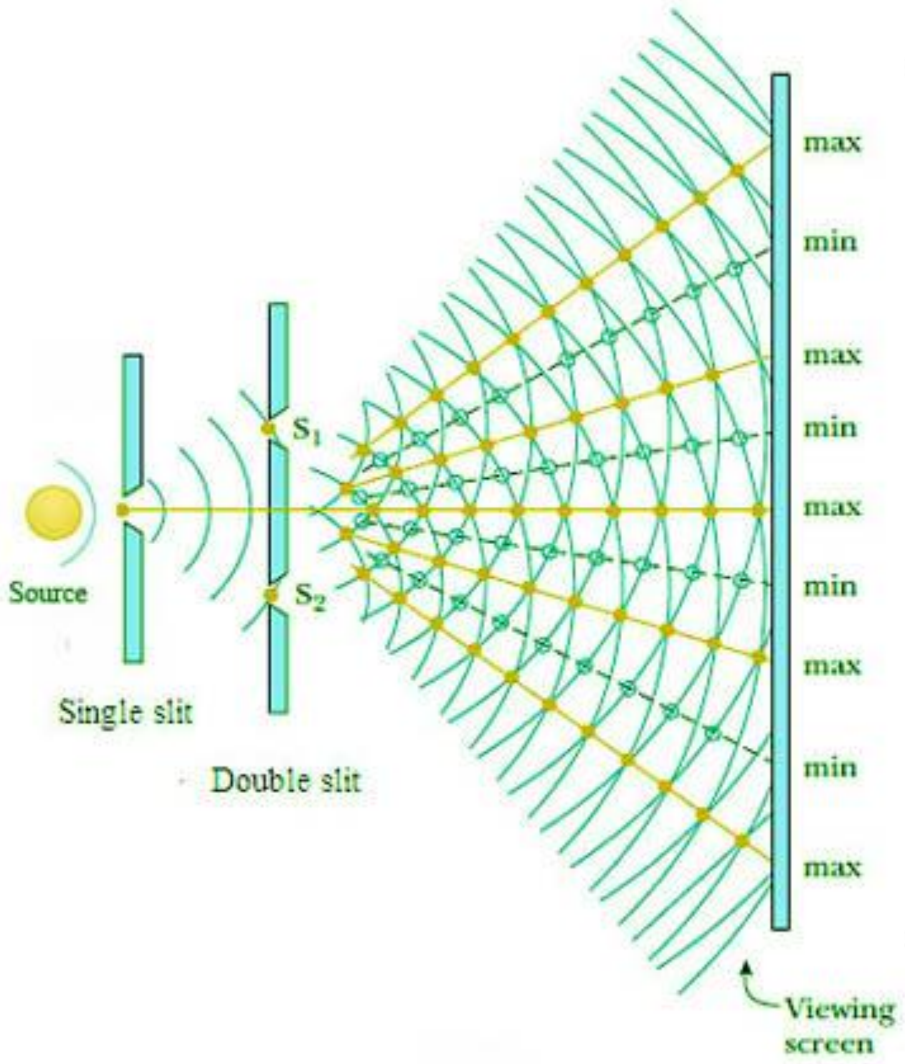
Effect of combining two rays
in same phase.

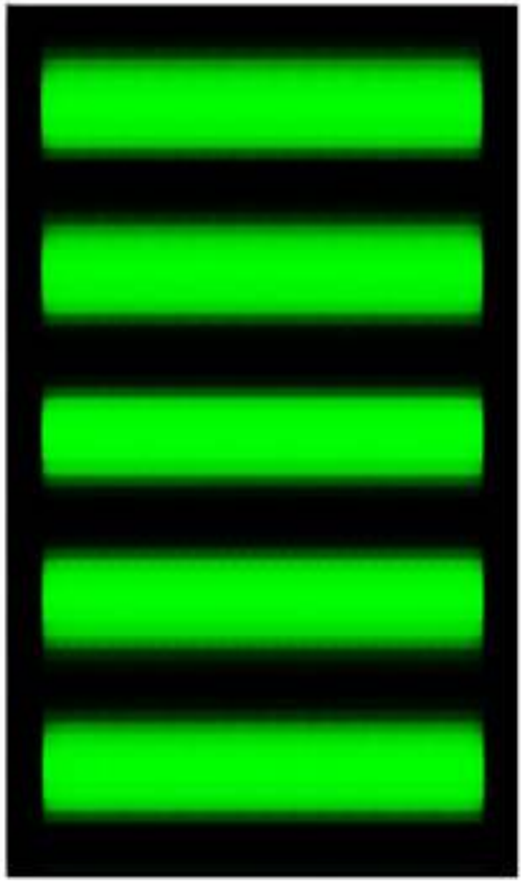
interference of two rays



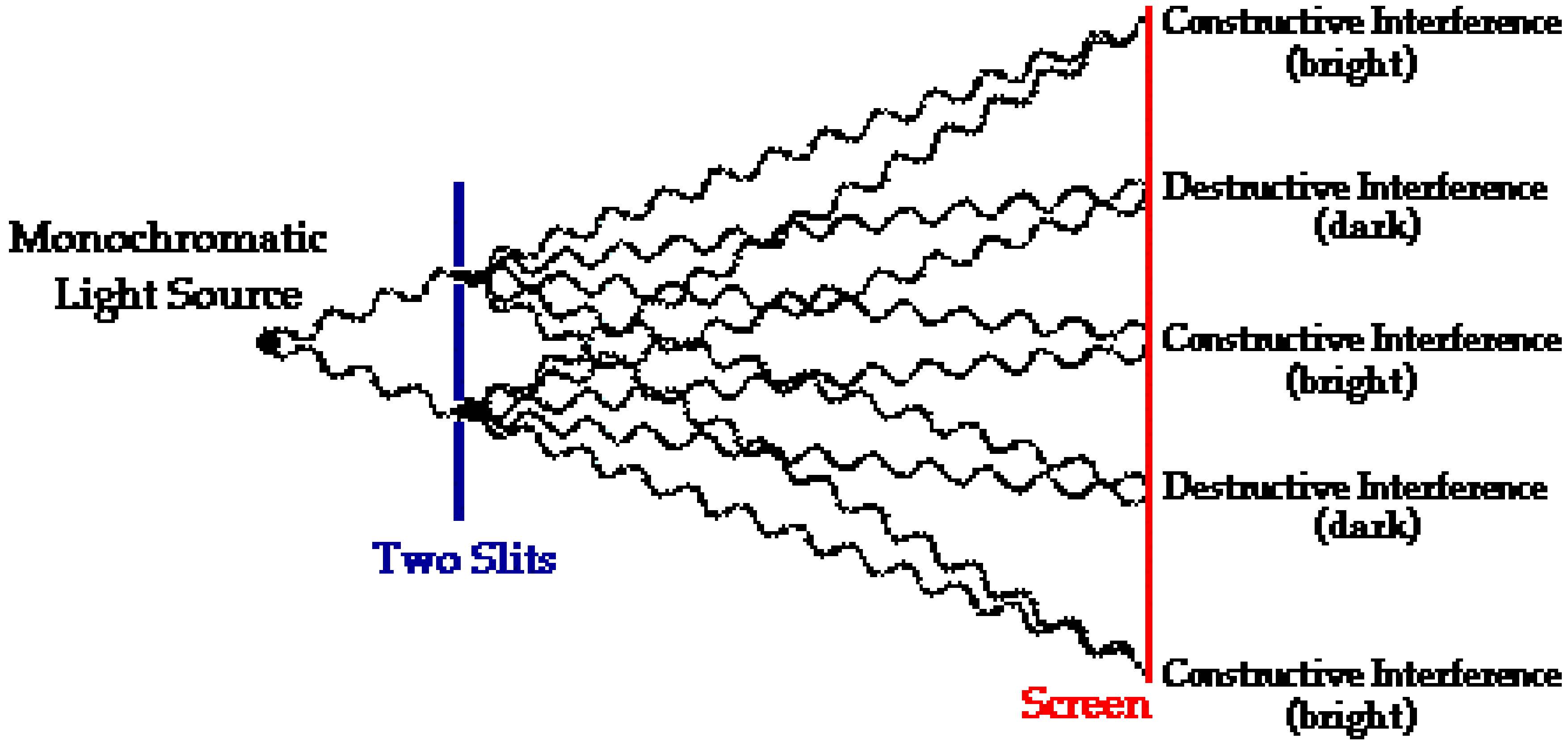
Effect of Combining two rays
out of phase by λ/2

Production of interference bands





Interference pattern on screen

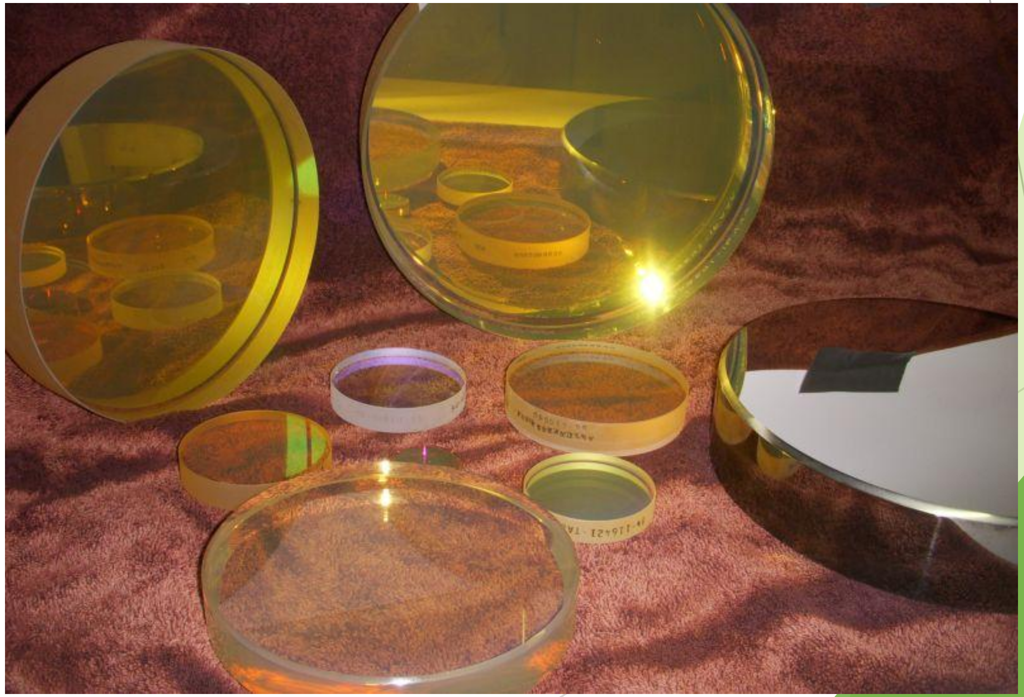


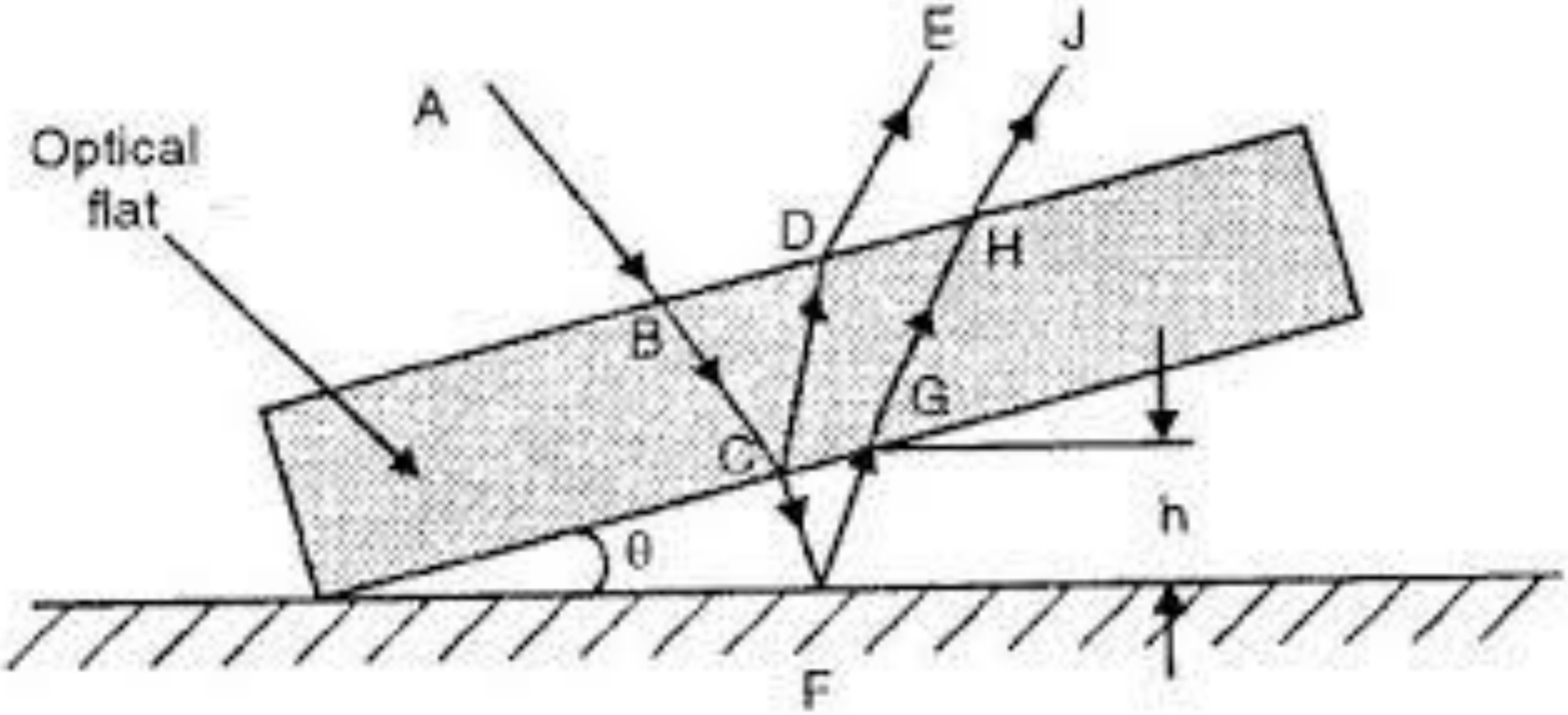
A two-point source interference pattern creates an alternating pattern of bright and dark lines when it is projected onto a screen.

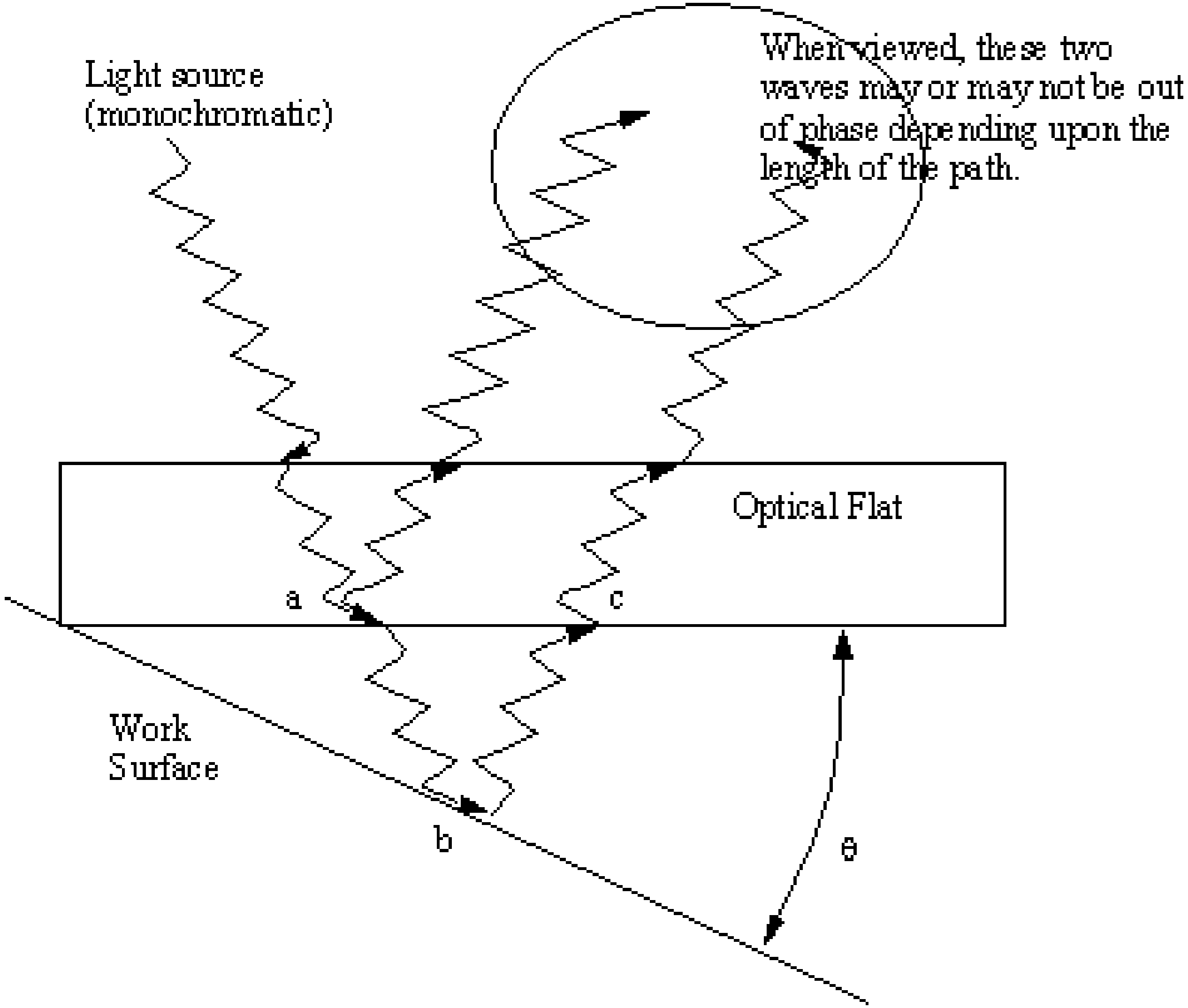
Interferometry applied to flatness testing

Optical flat

- Circular piece of optical glass or fused quartz having two plane faces flat and parallel
- High wear quality
- Temperature resistant

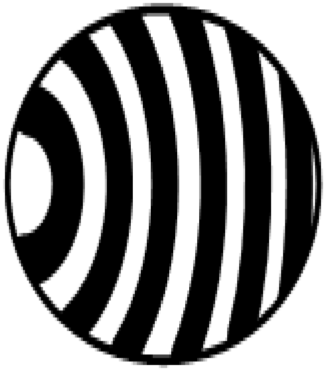


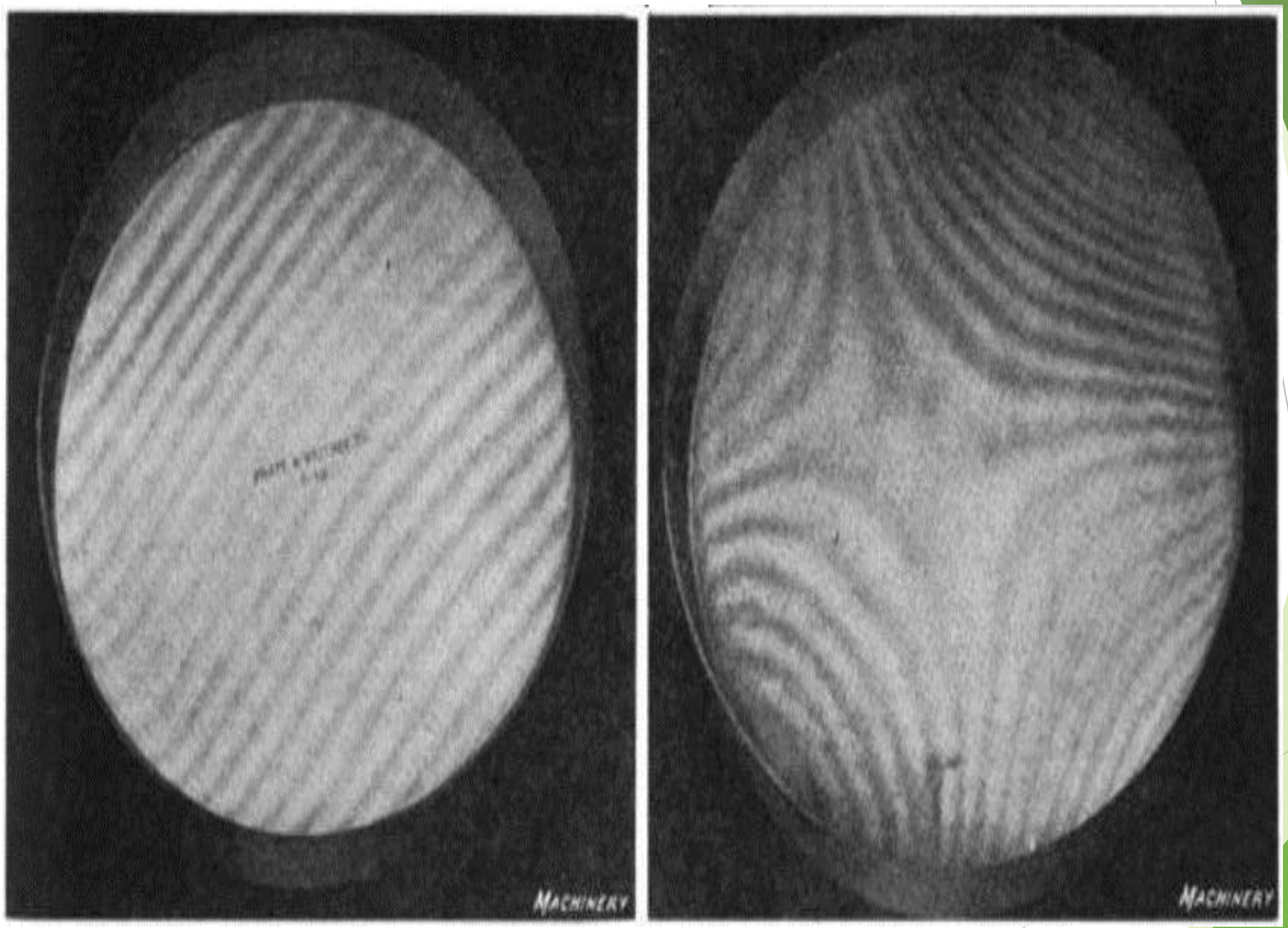

Optical
flat
A
B
C
D
E
J
H
G
h
θ
F


Light source
(monochromatic)
When viewed, these two
waves may or may not be out
of phase depending upon the
length of the path.
Optical Flat
a
c
Work
Surface
b
θ

Good

Poor

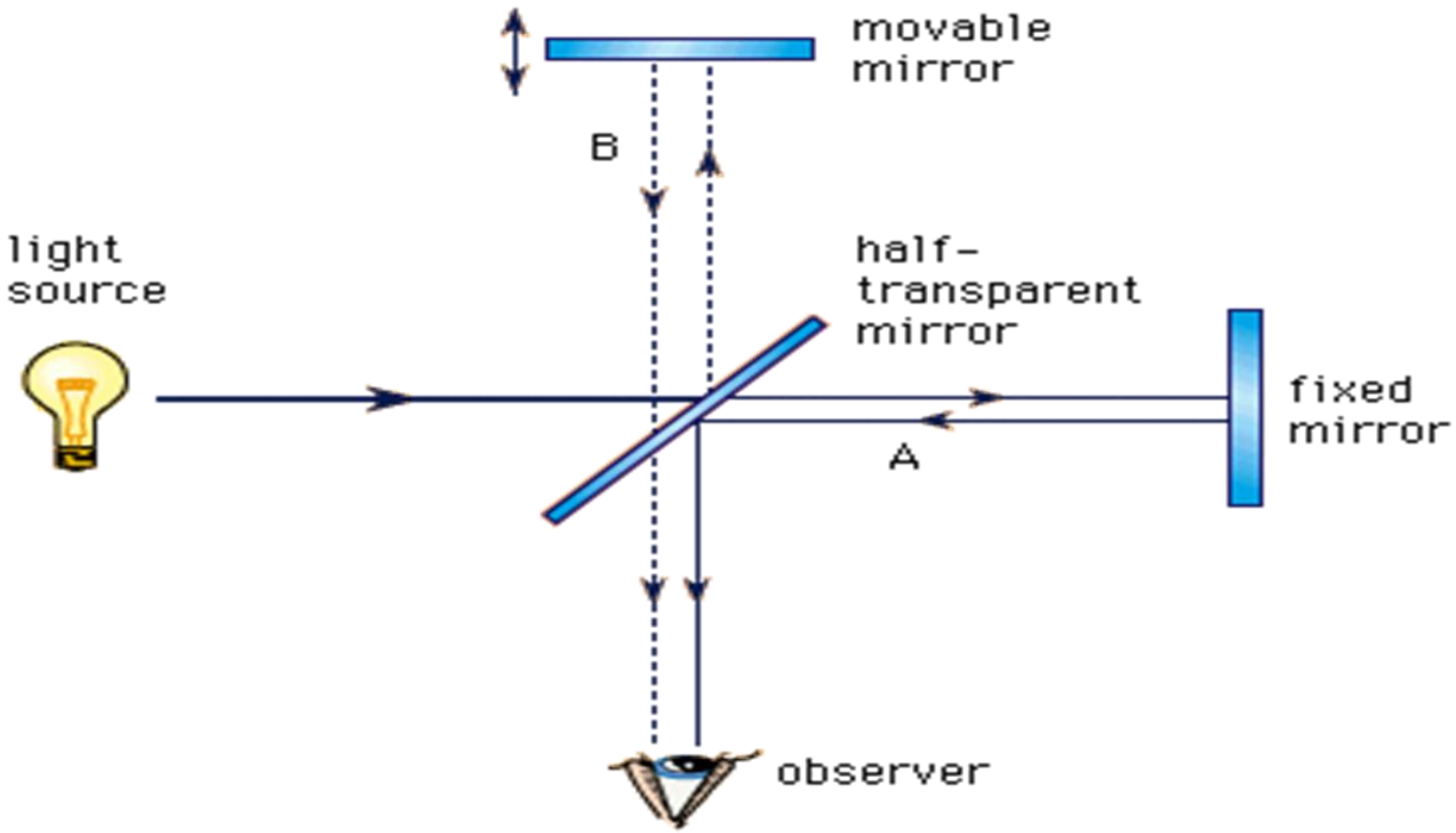




Interferometer

- Interferometers are optical instruments used for measuring flatness
- Determining minute differences in length by direct reference to the wavelength of light

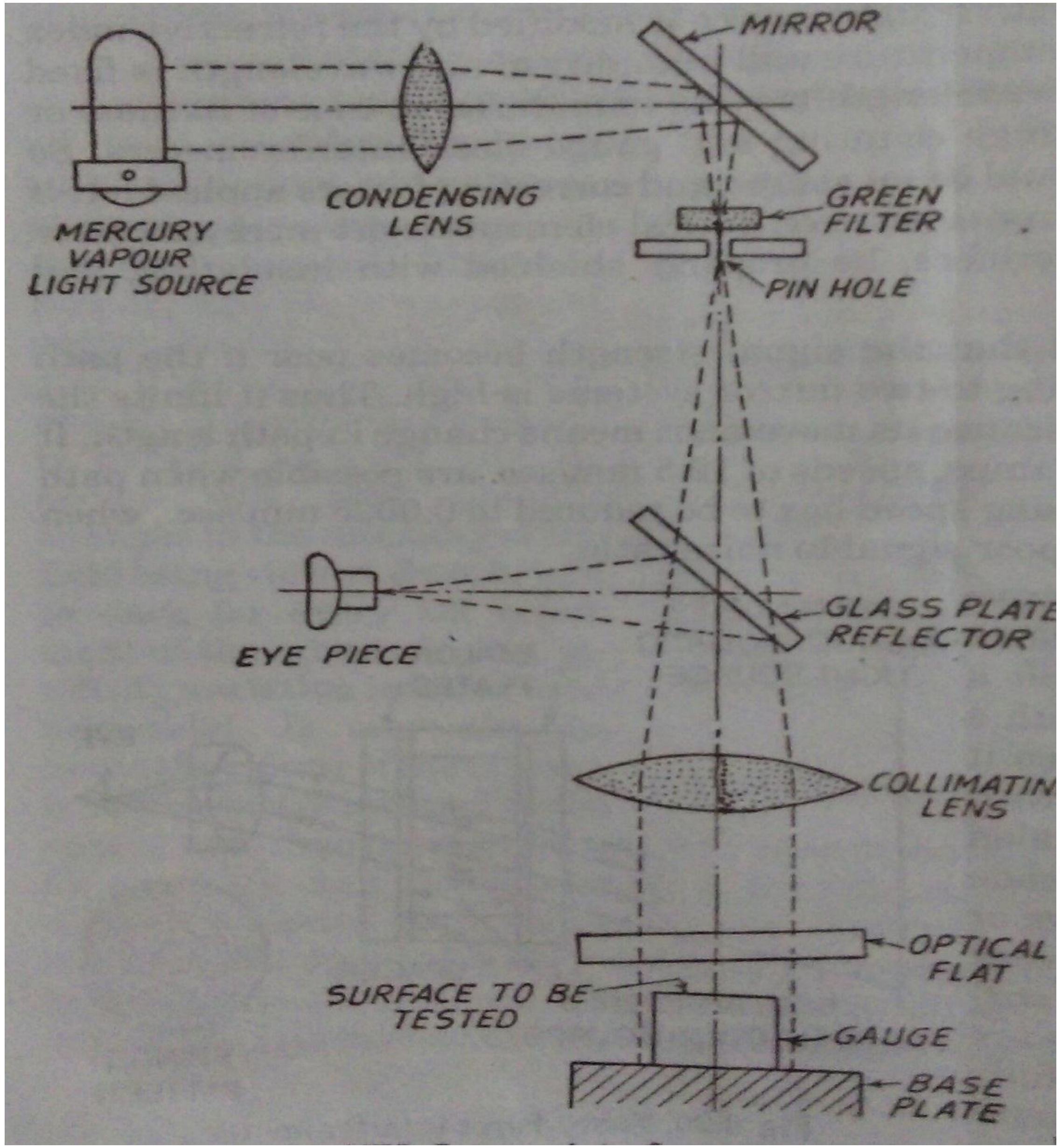
Interferometer



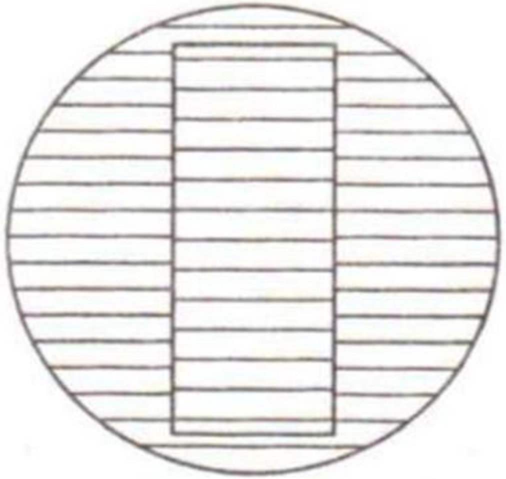
N.P.L Flatness Interferometer

- designed by National Physical Laboratory
- mainly used for checking the flatness of flat surfaces

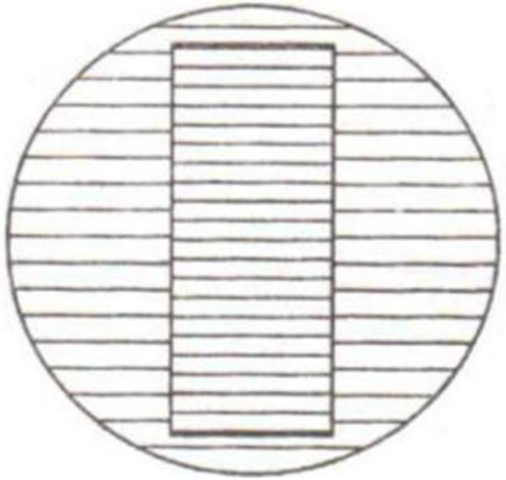
N.P.L Flatness Interferometer



Fringes

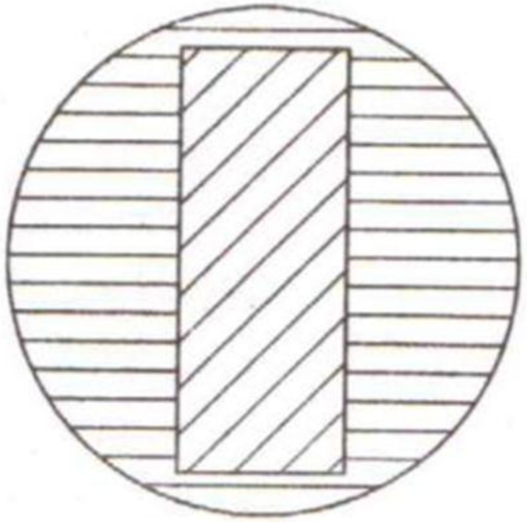


Gauge face flat and parallel to base plate

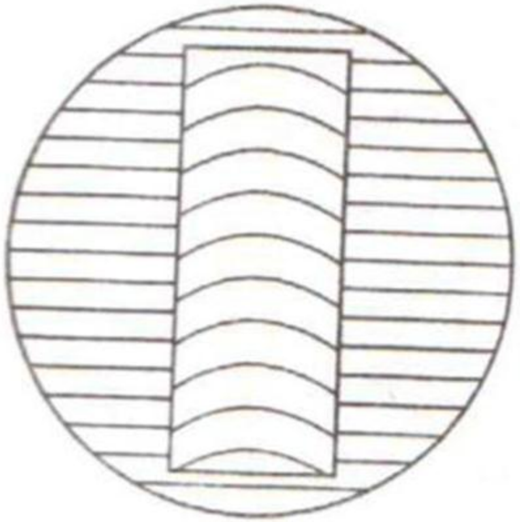


Gauge face flat but not parallel to base plate

Fringes

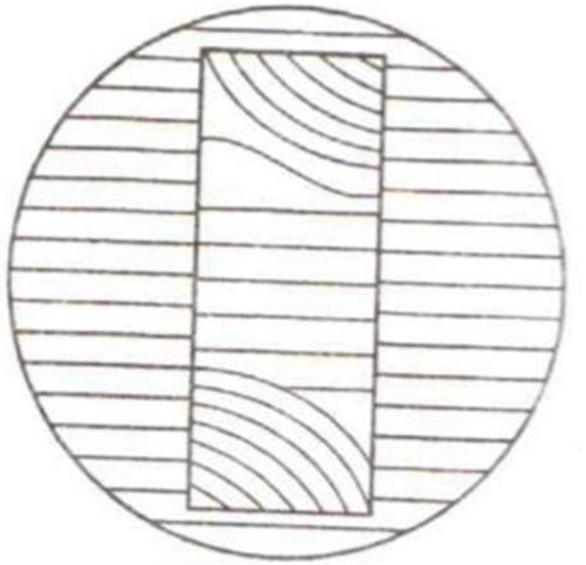


Surface of gauge
is inclined to base plate



Gauge surface
convex/concave

Fringes



Slight rounding off
at corners

Laser beam interferometer

- Laser interferometer uses A.C. laser as the light source and thus enables the measurements to be made over longer distance
- It must be understood that white light emitted by a lamp is combination of waves at different frequencies but laser generates a continuous train of light waves, resulting into high coherence.

Advantages

- high repeatability and resolution of displacement measurement (0.1 µm)
- high accuracy
- long-range optical path (60 m)
- easy installation
- no change in performance due to ageing or wear and tear

Laser beam interferometer- working